Towards Controllable and Personalized Review Generation

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Problem

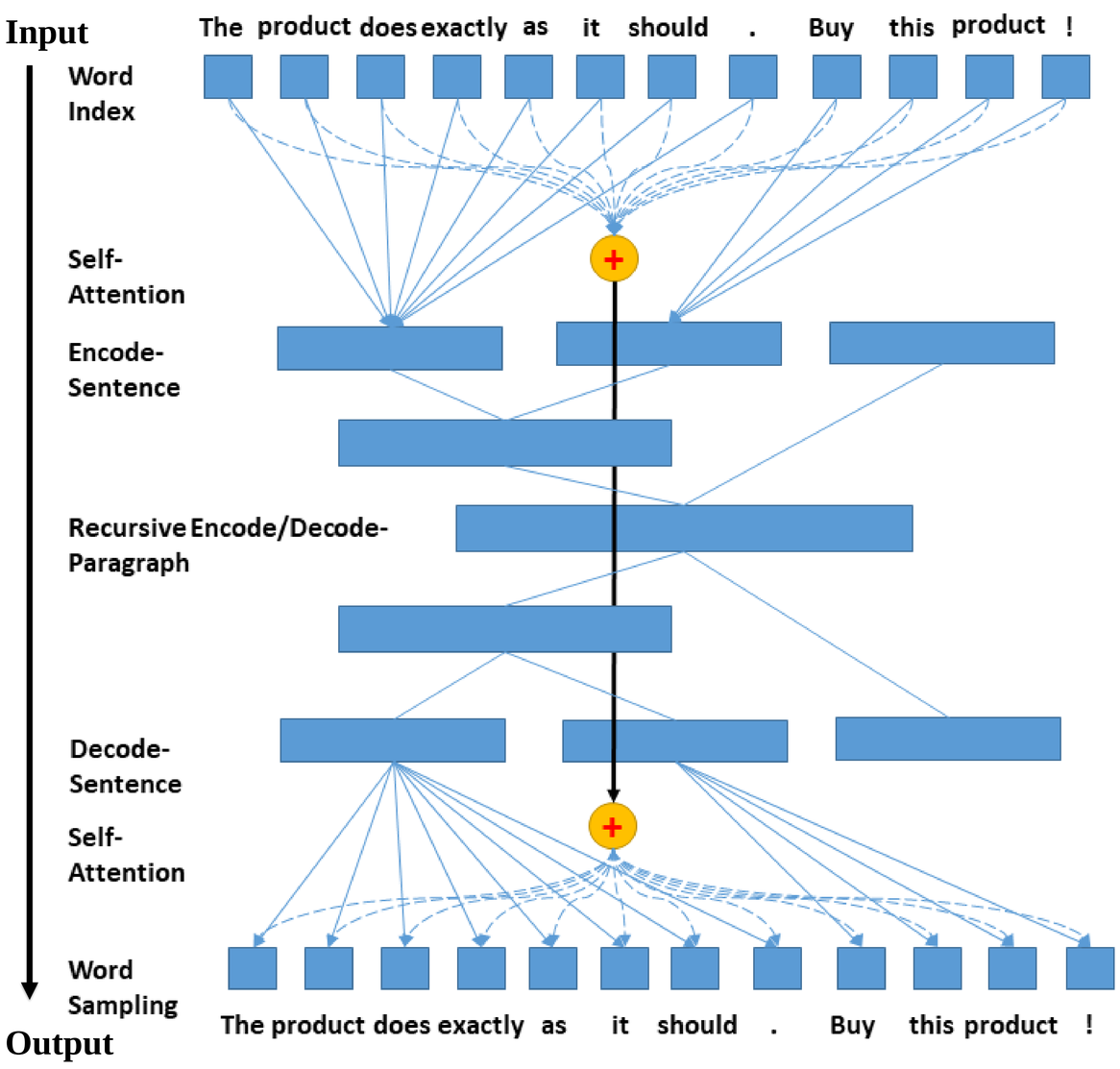
Previous models did not

- consider the inner hierarchical word-sentence-paragraph structure within user reviews.
- address the problem of **controllable** and **personalized** review generation targeted to the specific product and user
- include production descriptions in the generation process

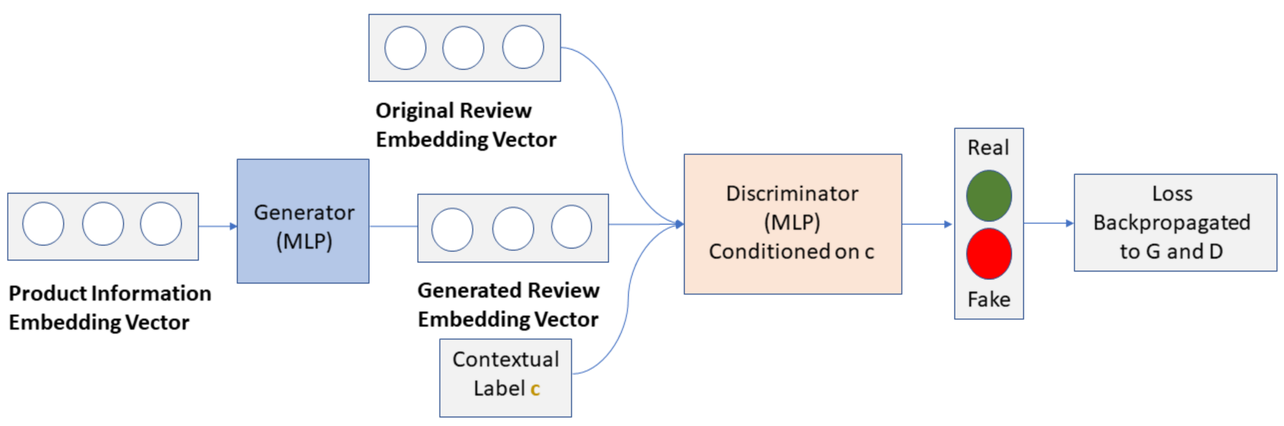
Solution: RevGAN

RevGAN consists of three components:

1. Self-Attentive Recursive Autoencoder that maps discrete user reviews and product descriptions into continuous embeddings



2. Conditional Discriminator that controls the sentiment of generated reviews



$$D(x|c) = \begin{cases} 1, \text{organic reviews matching label c} \\ 0, \text{others} \end{cases}$$

3. Personalized Decoder that decodes the generated review embeddings based on personalized writing styles.

Writing Style refers to the user's distinctive vocabulary choices and style of expression in his review creations. We adjust the output distribution vector w accordingly:

$$W' = W * W_{style}$$

Experiments

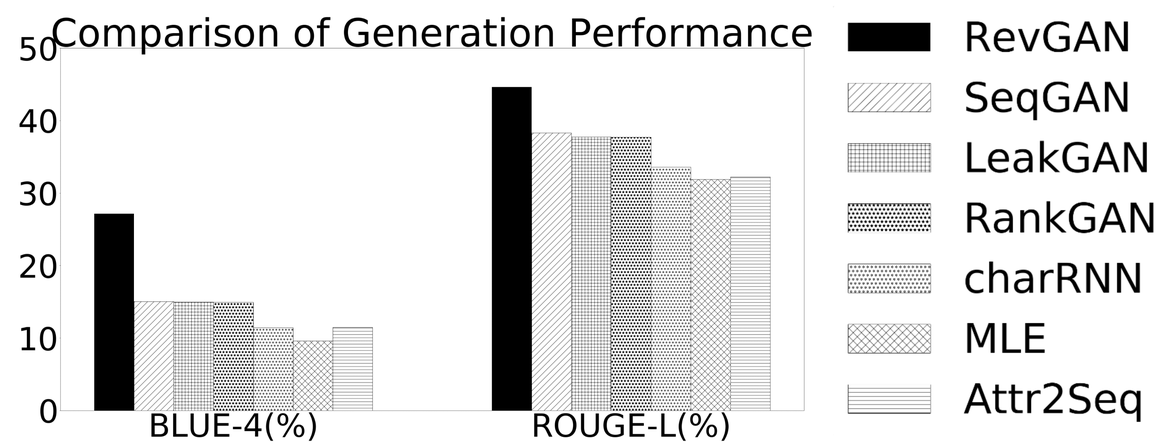
Datasets

- Amazon Review Dataset (Musical, Automotive, Patio)

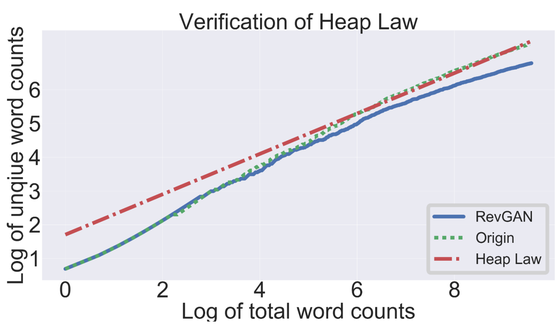
Baselines (compared with our RevGAN model)

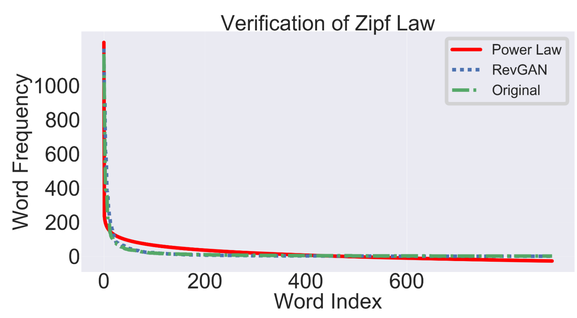
- SeqGAN
- LeakGAN
- RankGAN
- cGAN
- charRNN
- MLE
- Attr2Seq

Comparison of Generation Performance



Verification of Heap Law and Zipf Law





Confusion Matrix of Empirical Test

We ask participants to label whether they believe the given review is written by machines or humans

Empirical Label	Actual Value	
	Origin	RevGAN
Human	119	61
Machine	118	62

Conclusions

- **Self-Attention mechanism helps to improve review generation quality.**
- **Personalization plays an important role in deciding generation performance**
- **Conditional discriminator enables us to control for sentiment of output reviews.**